

COMPUTATIONAL HADRON SPECTROSCOPY

Mass Spectra and Decay Widths of Heavy Mesons via *the Gaussian Expansion Method*

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FOUNDATIONS

■ Introduction

Heavy quarkonia — a charm (c) or bottom (b) quark bound to its antiquark — are the “**hydrogen atom of QCD**.” Their large constituent masses slow the quarks ($v^2/c^2 \ll 1$), so a **non-relativistic Schrödinger** equation with a single phenomenological potential already captures the measured spectrum and the electromagnetic transitions between levels.

The problem. A confining potential cannot be derived from QCD in closed form; its shape and the size of relativistic spin corrections must be *fixed by data*, then tested for predictive power on states and widths not used in the fit.

This work. From *one* Cornell potential per sector we compute the full mass spectrum *and* the leptonic, two-photon and radiative decay widths — the same wavefunctions feed both, so a single fit faces two independent observable classes. Three sectors ($c\bar{c}$, $b\bar{b}$, $b\bar{c}$ (B_c)) are benchmarked against **PDG 2026** and six quark-model studies.

■ Full Hamiltonian

$$H = \frac{p^2}{2\mu} + V(r) + H_{SS} + H_{LS} + H_T$$

Kinetic + potential + Breit-Fermi spin-dependent terms

$$V(r) = -\frac{4}{3} \frac{\alpha_s}{r} + b r + c$$

Cornell: one-gluon Coulomb + linear confinement + offset

$$H_{SS} = \frac{32\pi\alpha_s}{9m_1m_2} \tilde{\delta}_{\sigma_0}^3(r) \mathbf{S}_1 \cdot \mathbf{S}_2$$

Spin-spin hyperfine: contact term smeared to width σ_0

$$H_{LS} = \frac{1}{2m_1m_2} \left(\frac{1}{r} \frac{dV_V}{dr} - \frac{1}{3r} \frac{dV_S}{dr} \right) \mathbf{L} \cdot \mathbf{S}$$

Spin-orbit: fine structure of P and D waves

$$H_T = \frac{1}{6m_1m_2} \left(\frac{1}{r} \frac{dV_V}{dr} - \frac{d^2V_V}{dr^2} \right) S_{12}$$

Tensor: mixes 3S_1 - 3D_1 ; $S_{12} = 6(\mathbf{S}_1 \cdot \hat{r})(\mathbf{S}_2 \cdot \hat{r}) - 2\mathbf{S}_1 \cdot \mathbf{S}_2$

■ Variational Method (GEM)

Each radial wavefunction is expanded in a fixed ladder of Gaussians, turning the differential eigenvalue problem into a small generalized *matrix* eigenproblem solved once per state.

$$u_{nl}(r) = \sum_{i=1}^N c_i r^{l+1} e^{-\nu_i r^2}, \quad \nu_i = \nu_1 a^{i-1}$$

Geometric Gaussian basis ($N = 25$) $\Rightarrow$ generalized eigenproblem $Hc = ESc$

- All matrix elements integrated **analytically** from closed-form Γ -function integrals — no quadrature, no $1/r$ singularity.
- Geometric progression spans the compact Coulombic region and the diffuse linear confinement tail in one basis.
- Convergence from the virial ratio $\langle r \cdot \nabla V \rangle / 2\langle T \rangle \approx 1.000$.
- 3S_1 - 3D_1 tensor mixing diagonalized **exactly** — critical for $\psi(3770)$ and $\psi(4160)$.

■ Hypervirial Theorem for $|R(0)|^2$

$$|R_S(0)|^2 = 2\mu \left[\frac{4\alpha_s}{3} \langle r^{-2} \rangle + b \right]$$

Wavefunction at the origin as a well-posed expectation value — never extrapolated to $r = 0$. Every S -wave annihilation width feeds from it.

■ Fitted Cornell Parameters

SYM	PARAMETER	$c\bar{c}$	$b\bar{b}$
α_s	Strong coupling	0.5118 ± 0.0209	0.3358 ± 0.0093
b [GeV ²]	String tension	0.1342 ± 0.0048	0.1960 ± 0.0036
c [GeV]	Energy offset	-0.023 ± 0.027	$+0.011 \pm 0.014$
m_q [GeV]	Contact smearing	1.224 ± 0.460	3.444 ± 1.590
m_q [GeV]	Quark mass (fixed)	1.500	4.730

B_c : α_s, σ_0 log-interpolated from $c\bar{c}/b\bar{b}$; b, c fitted to the two known B_c masses. Uncertainties via finite-difference propagation of the Cornell covariance.

KEY REFERENCES

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- [4] W. Lucha, F. Schöberl, D. Gromes, *Phys. Rep.* **200**, 127 (1991) — Breit-Fermi.
- [5] R. Van Royen, V. F. Weisskopf, *Nuovo Cim. A* **50**, 617 (1967) — leptonic width.
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- [8] N. R. Soni *et al.*, *Eur. Phys. J. C* **78**, 592 (2018) — Cornell NR.
- [9] E. B. Gregory *et al.* (HPQCD), *Phys. Rev. D* **83**, 014506 (2011) — B_c^* lattice.

DECAY WIDTH FORMULAS

■ Annihilation & Radiative

$$\Gamma(^3S_1 \rightarrow e^+e^-) = \frac{4\alpha^2 e_q^2}{M^2} |R_S(0)|^2 \left(1 - \frac{16\alpha_s}{3\pi} \right)$$

S -wave leptonic (Van Royen-Weisskopf): $\Upsilon(nS), J/\psi, \psi'$

$$\Gamma(^3D_1 \rightarrow e^+e^-) = \frac{25\alpha^2 e_q^2}{2m_q^4 M^2} |R_D''(0)|^2 \left(1 - \frac{16\alpha_s}{3\pi} \right)$$

D -wave leptonic: $\psi(3770), \psi(4160) - R(0)=0$, so the rate is set by $|R_D''(0)|^2$ and the 3S_1 admixture

$$\Gamma(^1S_0 \rightarrow \gamma\gamma) = \frac{12\alpha^2 e_q^4}{M^2} |R_S(0)|^2$$

Pseudoscalar two-photon: η_c, η_b — same $|R(0)|^2$

$$\Gamma(^3P_0 \rightarrow \gamma\gamma) = \frac{27\alpha^2 e_q^4}{2m_q^4} |R_P'(0)|^2$$

Scalar χ_0 (Barbieri-Gatto-Remiddi 1976); $J=1$ Yang-forbidden

$$\Gamma(^3P_2 \rightarrow \gamma\gamma) = \frac{18\alpha^2 e_q^4}{5m_q^4} |R_P'(0)|^2$$

Tensor χ_2 ; both P -wave rates use $|R_P'(0)|^2$ from the GEM basis

$$\Gamma(M1) = \frac{4\alpha e_q^2}{3m_q^2} k^3 |\langle f | j_0(kr/2) | i \rangle|^2$$

Magnetic dipole $V \rightarrow P\gamma$; $k = (M_i^2 - M_f^2)/2M_i$

$$\Gamma(E1) = \frac{4\alpha e_q^2}{9} k^3 (2J_f + 1) |\langle f | r | i \rangle|^2$$

Electric dipole $P \rightarrow S\gamma$; $\langle r \rangle$ from analytic GEM integrals

GOODNESS OF FIT

$b\bar{b}$ RMS mass deviation

8.9 MeV

$c\bar{c}$ RMS mass deviation

29.8 MeV

Global χ^2/dof (masses)

0.54

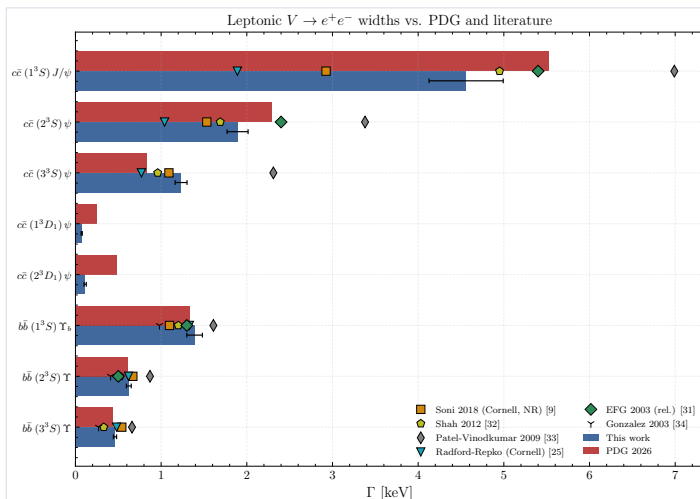
Global χ^2/dof ($+e^+e^-$)

1.12

KEY TENSION · $\psi(3770)$ The 1^3D_1 leptonic width is 10.3σ below PDG. A pure D -wave has $R(0)=0$; its e^+e^- rate comes entirely from the 3S_1 admixture set by $\langle 2^3S | V_{\gamma} | 1^3D \rangle$. Single-channel tensor mixing is weaker than the physical coupled-channel mixing near the open-charm $D\bar{D}$ threshold — a loop effect outside any potential model.

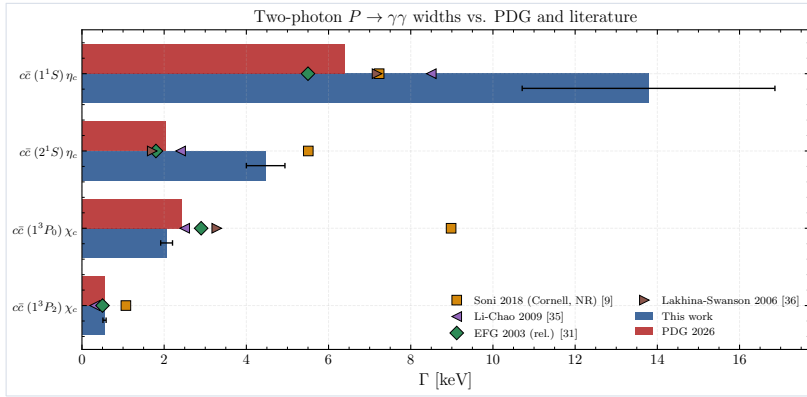
RESULTS · DECAYS

■ Leptonic $V \rightarrow e^+e^-$



$V \rightarrow e^+e^-$ widths. $b\bar{b}$ S -waves sit sub- 1σ ; $c\bar{c}$ rates run 2– 3σ low (NR limit); the D -wave $\psi(3770)$ falls 10.3σ short (tension box above).

■ Two-photon $P \rightarrow \gamma\gamma$



$P \rightarrow \gamma\gamma$. The η_c overshoot ($\times 2.2$) is the expected leading-order NR excess; $\chi_{c2} \rightarrow \gamma\gamma$ matches PDG to 2%.

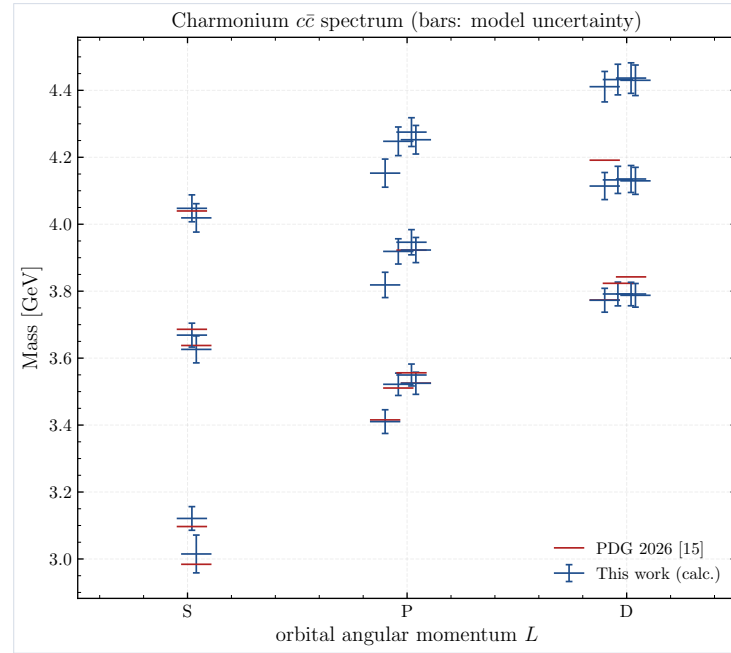
■ Selected decay widths

CHANNEL	Γ_{calc} [KEV]	Γ_{PDG}	PULL
$J/\psi \rightarrow e^+e^-$	4.560 ± 0.433	5.528	-2.2σ
$\psi(2S) \rightarrow e^+e^-$	1.893 ± 0.123	2.293	-3.0σ
$\Upsilon(1S) \rightarrow e^+e^-$	1.391 ± 0.091	1.340	$+0.5\sigma$
$\Upsilon(2S) \rightarrow e^+e^-$	0.623 ± 0.029	0.612	$+0.4\sigma$
$\eta_c \rightarrow \gamma\gamma$	13.79 ± 3.07	6.39	$+2.4\sigma$
$\chi_{c0} \rightarrow \gamma\gamma$	2.060 ± 0.144	2.433	-2.1σ
$\chi_{c2} \rightarrow \gamma\gamma$ ★	0.549 ± 0.038	0.559	-0.2σ
$\chi_{c1} \rightarrow J/\psi \gamma$ [E1]	350.9 ± 17.7	—	—
$\Upsilon(1S) \rightarrow \eta_b \gamma$ [M1]	0.012 ± 0.014	—	—

$\text{Pull} = (\Gamma_{\text{calc}} - \Gamma_{\text{PDG}}) / (\sigma_{\text{exp}} \oplus \sigma_{\text{comp}})$. ★ excellent agreement.

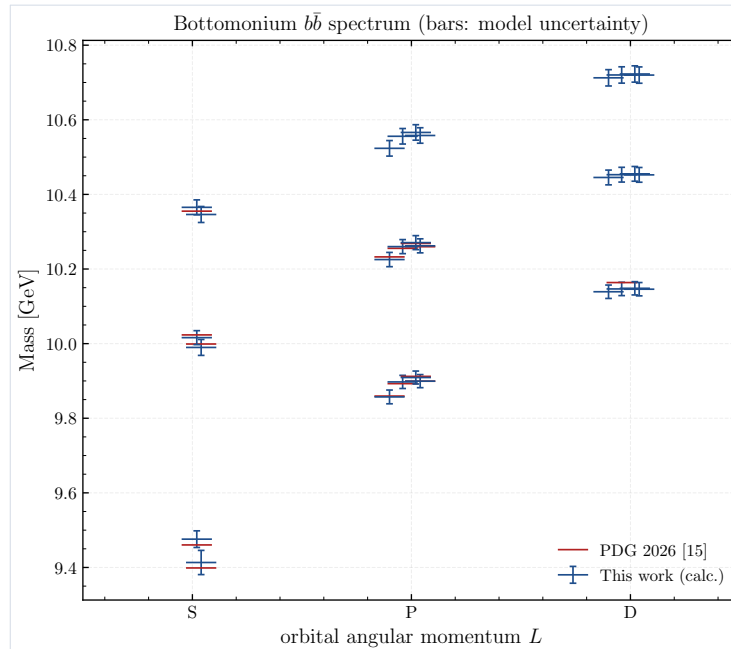
RESULTS · MASS SPECTRA

■ Charmonium $c\bar{c}$



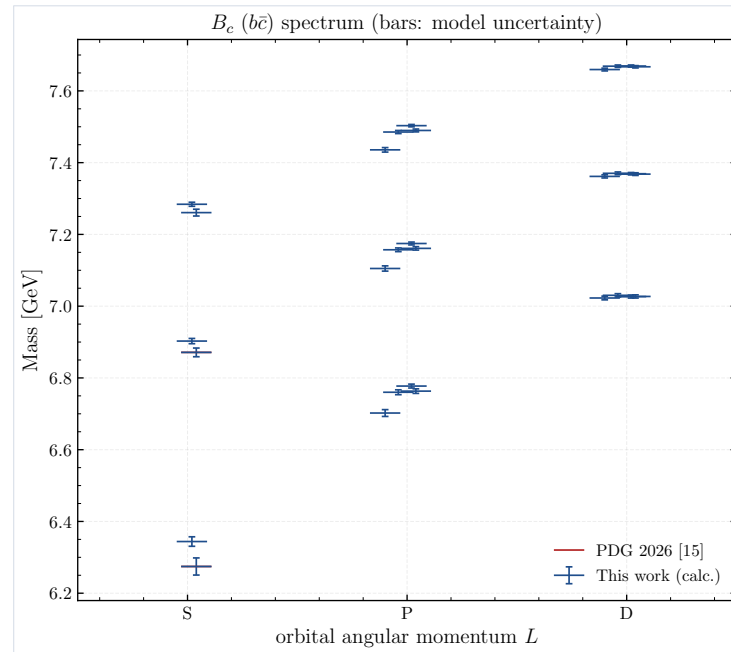
$c\bar{c}$ vs PDG 2026. Blue = this work (model-uncertainty bars); red = PDG. RMS = 29.8 MeV over 14 states; largest tension at 2^3D_1 (-77 MeV, near threshold).

■ Bottomonium $b\bar{b}$



$b\bar{b}$ vs PDG 2026. Heavier quarks $\Rightarrow$ better NR limit $\Rightarrow$ RMS drops to **8.9 MeV**; all pulls sub- 1σ .

■ B_c meson $b\bar{c}$



B_c spectrum. Two known masses anchor the fit (filled); all triplet and excited levels are **predictions** — the α_s, σ_0 are RG log-interpolated from $c\bar{c}/b\bar{b}$.

■ Selected mass comparison

STATE	M_{calc}	M_{PDG}	Δ	PULL
CHARMONIUM $c\bar{c}$				
J/ψ (1^3S_1)	3121 ± 35	3096.9	$+24.2$	$+0.68\sigma$
η_c (1^1S_0)	3015 ± 57	2984.1	$+30.9$	$+0.55\sigma$
h_c (1^1P_1) ★	3525 ± 33	3525.4	-0.5	-0.02σ
ψ (1^3D_1) ★	3773 ± 36	3773.7	-0.8	-0.02σ
BOTTOMONIUM $b\bar{b}$				
Υ (1^3S_1)	9476 ± 22	9460.3	$+15.5$	$+0.69\sigma$
η_b (1^1S_0)	9413 ± 32	9398.7	$+14.6$	$+0.45\sigma$
h_b (1^1P_1) ★	9900 ± 18	9899.3	$+0.2$	$+0.01\sigma$
B_c MESON $b\bar{c}$ (FIT ANCHORS)				
B_c (1^1S_0) ★	6275 ± 24	6274.5	0.0	0.0 σ
B_c (2^1S_0)	6871 ± 12	6871.2	0.0	0.0 σ

Masses in MeV. ★ sub-MeV agreement. Pull = Δ/σ , $\sigma = \sigma_{\text{exp}} \oplus \sigma_{\text{comp}}$ (propagated Cornell covariance, $\sigma_{\text{comp}} \approx 20$ MeV $b\bar{b}$, 40 MeV $c\bar{c}$); all sub- 1σ .

PREDICTIONS · UNMEASURED STATES

■ Key predictions

STATE	M_{calc} [MEV]	ANCHOR
B_c^* (1^3S_1) ★	6344 ± 13	lattice / lit.
B_c^* (2^3S_1) ★	6903 ± 7	consistent
η_c (3^1S_0)	4019 ± 42	predicted
ψ (2^3D_1)	4114 ± 41	PDG 4191?
Υ (1^3D_1)	10139 ± 18	predicted

Anchors: B_c^* vs. HPQCD lattice [9]; $\psi(2^3D_1)$ is -1.9σ from the tentative PDG $\psi(4191)$ assignment. ≥ 1 prediction per sector.

CONCLUSION A single Cornell potential per sector reproduces the PDG masses — **RMS 8.9/29.8 MeV** for $b\bar{b}/c\bar{c}$ — and the S -wave leptonic widths from the same wavefunctions; the residual $\psi(3770)$ and $\eta_c \rightarrow \gamma\gamma$ tensions mark precisely where the non-relativistic single-channel picture ends.

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Full report & code

levantkaanoguz.com/400/400_final_report.pdf

